

Nada Mohamed

AI/ML Engineer · Blockchain Engineer

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SUMMARY

Computer Science graduate (2026) building at the intersection of **machine learning** and **distributed ledgers**. Fine-tuned LLMs (DeepSeek-Coder, FLAN-UL2) for a multimodal agent, shipped deep-learning, computer-vision and reinforcement-learning systems, and engineered **Hyperledger Fabric** chaincode in Go for a regulated payroll platform validated by 91 passing smart-contract tests. Completed Microsoft's 7-month ML Engineering program (DEPI); First Place, AI/ML Project Submission at AASTMT.

TECHNICAL SKILLS

AI / ML: PyTorch, TensorFlow/Keras, Hugging Face Transformers, Scikit-learn, OpenCV, Whisper, Librosa, NumPy, Pandas · LLM fine-tuning & instruction tuning, agent orchestration, NLP, computer vision, speech, reinforcement learning, explainable AI (SHAP, LIME, Grad-CAM)

Blockchain & Security: Hyperledger Fabric, chaincode (Go), RAFT ordering, endorsement policies, MPC / threshold key custody (Privy), non-custodial wallets, X.509 PKI, applied cryptography, STRIDE threat modelling

Languages: Python, Go, TypeScript, JavaScript, C/C++, Java, SQL, Bash, R

Serving & Infra: FastAPI, Flask, React, Docker, Kubernetes, AWS (RDS), MySQL, SQLite, Git, GitHub Actions CI/CD, Linux, Pytest

EXPERIENCE

AI/ML Engineer Intern — *DEPI (Microsoft ML Engineering curriculum)*

Nov 2024 – May 2025 · Cairo

- Completed a 7-month intensive program spanning NLP, generative AI, computer vision and model deployment, shipping a hands-on project at each milestone.
- Delivered the end-to-end capstone, **Jarvis AI Assistant**: multimodal workflows (code, vision, voice) with intelligent agent orchestration.

Summer Intern — *Dell Technologies*

Aug 2025 – Sep 2025 · Cairo

- Completed engineering workshops on containerization (Docker, Kubernetes) and software engineering practice; led the Academy Project from concept to business proposal.

PROJECTS

Payramid — Blockchain Payroll Microservice | *Go, Hyperledger Fabric, Privy MPC, TypeScript*

Graduation Project · 2025 – 2026

- Built a payroll-orchestration and immutable-audit layer letting employers disburse salaries instantly with **cryptographically verifiable proofs of payment**, settling on Egypt's national rails (InstaPay, Meeza, ACH) within banking (Law 194/2020) and data-protection (Law 151/2020) regulation.
- Engineered the smart-contract layer as **Fabric chaincode in Go** on a 3-organization RAFT network with MAJORITY endorsement; validated by an automated suite of **91 passing tests** covering serialization integrity, double-spend prevention and a full STRIDE threat model.
- Added non-custodial wallets via **threshold (MPC) key management** and an encrypted SMS relay with one-time withdrawal codes for feature phones; authored the IEEE 830 SRS and IEEE 829 test plan.

Jarvis AI Assistant | *Python, PyTorch, Hugging Face, DeepSeek-Coder, FLAN-UL2, Whisper*

2025 – Present

- Fine-tuned DeepSeek-Coder and FLAN-UL2** with prompt-based instruction tuning for multimodal encoder-decoder workflows: code generation, image captioning and voice interaction.
- Integrated Whisper speech recognition and agent orchestration with custom toolchains for autonomous task execution; packaged as an offline desktop app (React, Flask, PyWebView).

Music Genre Detection | *TensorFlow/Keras, Librosa, Flask*

- Designed a **CNN-LSTM-Attention** model classifying mel-spectrograms into 10 genres, served via Flask, with explainability (SHAP, LIME, Grad-CAM, Integrated Gradients), unit tests and Sphinx docs.

CMASS Analytics Assistant | *Python, FastAPI, Groq Llama 3.3 70B*

- Built a NASA turbofan-degradation service computing **Remaining Useful Life** and ranking sensors by informativeness, with an LLM-powered natural-language QA endpoint over failure signals.

Oribu — Facial Recognition | *Python, OpenCV, Siamese NN*

First Place, AI/ML Project · AASTMT

- Built one-shot face verification with a **Siamese network and contrastive loss**, running real-time recognition with automated Twilio SMS alerts.

Simple Certificate Authority | *Python, Cryptography, X.509*

- Built a modular X.509 CA: root-CA init, issuance, revocation (CRL), verification, renewal and PKCS#12 export, with passphrase-encrypted keys and atomic writes.

EDUCATION

B.Sc. Computer Science — *Arab Academy for Science, Technology & Maritime Transport (dual degree, Univ. of Northampton)*

2022 – 2026

GPA 3.67 / 4.0 · Coursework: Machine Learning, Computer Vision, NLP, Algorithms & Optimization, Parallel Computing, Databases

CERTIFICATIONS & LANGUAGES

Microsoft ML Engineer Course (DEPI) · AI & Machine Learning, Universitat Autònoma de Barcelona · Computer Science Camp, Carnegie Mellon University

Languages: Arabic (Native), English (Fluent), Spanish (Intermediate)